

1. Waveshare setup

1.1. Install waveshare

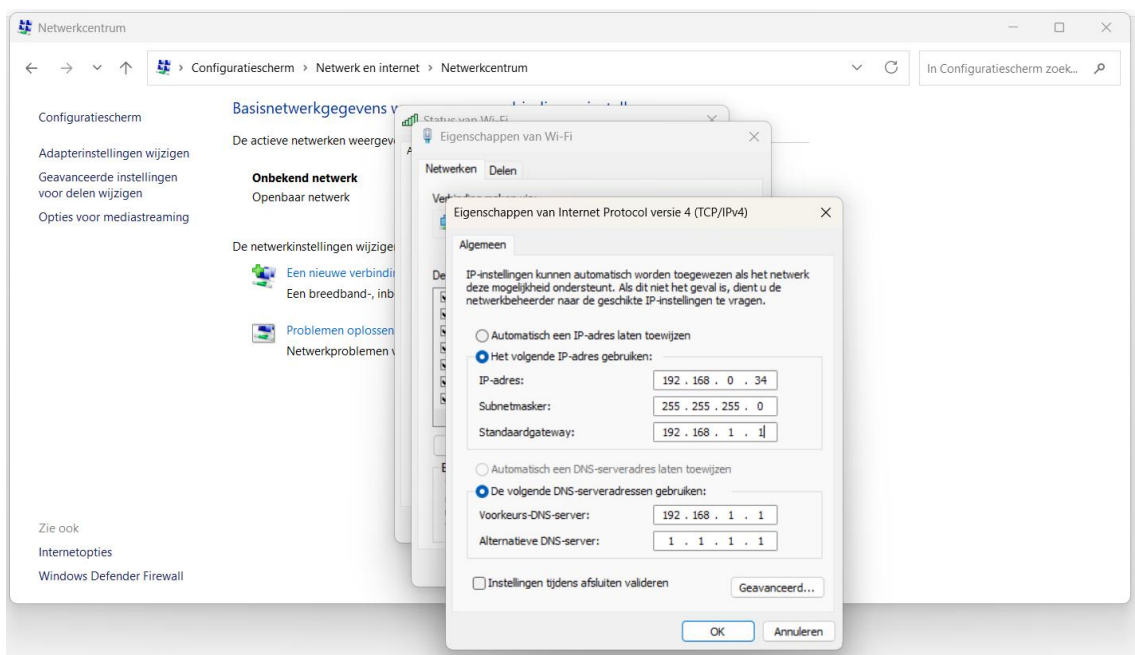
- A. Use a 6v to 36v DC adapter
- B. Connect the DC adapter to the V+ and V- ports of the waveshare
- C. Connect a UTP cable to the RS485_A3 and RS485_B3 ports of the inverter and the 485A and 485B ports of the waveshare.

1.2. Reset waveshare

- A. Bridge the GND and DEF ports for 5 seconds

1.3. Access waveshare device

- A. Identify the default gateway of the home network
- B. Identify the ip address of the Waveshare device
- C. Check whether the Waveshare is on the same subnet
 - a. Check whether the default ip address of the Waveshare is not in use. In general Waveshare uses 192.168.1.254 or 192.168.1.200. If another device already has the waveshares same IP address, disconnect this device first.
 - b. If the subnet is different, adjust the ipv4 of the computer to match the waveshares subnet. Make sure to use an available ip address



Network 3

Network profile

☒ Public

Your PC is hidden from other devices on the network and can't be used for printer and file sharing.

☐ Private

For a network you trust, such as at home or work. Your PC is discoverable and can be used for printer and file sharing if you set it up.

[Configure firewall and security settings](#)

Metered connection

If you have a limited data plan and want more control over data usage, make this connection a metered network. Some apps might work differently to reduce data usage when you're connected to this network.

Set as metered connection

☐ Off

If you set a data limit, Windows will set the metered connection setting for you to help you stay under your limit.

[Set a data limit to help control data usage on this network](#)

IP settings

IP assignment:

Automatic (DHCP)

[Edit](#)

Edit IP settings

Manual

IPv4

☒ On

IP address

Subnet prefix length

Gateway

Preferred DNS

Alternate DNS

IPv6

Save

Cancel

Subnet prefix length =24

- D. If this doesn't work, Check the following
 - a. Subnet mask should be 255.255.255.0 (check CMD ipconfig)
 - b. Gateway IP in general 192.168.1.1 (check CMD ipconfig)
 - c. Check whether the network type is private
 - d. Temporarily disable the firewall
- E. If this doesn't work, connect the waveshare directly to the computer
 - a. Disable wifi
 - b. Connect PC and waveshare using an ethernet cable
 - c. Open vircom
 - d. Adjust IP, subnet mask and gateway to match your network

Device Settings

Device Info

Virtual Serial: Not Use
Dev Type:
Dev Name: WSDEV0001
Dev ID: 2877FE5529D4 [-]
Firmware Ver: V1.452

Function of the device

☐ Web Download
☒ DNS System
☒ REAL_COM Protocol
☒ Modbus TCP To RTU
☒ Serial Commnad
☒ DHCP Support
☐ Storage Extend
☒ Multi-TCP Connection

Network

IP Mode: Static
IP Address: 192 . 168 . 1 . 200
Port: 502
Work Mode: TCP Server
Net Mask: 255 . 255 . 255 . 0
Gateway: 192 . 168 . 1 . 1
Dest. IP/Domain: 192.168.1.200 Local IP
Dest. Port: 502

Serial

Baud Rate: 2400
Data Bits: 8
Parity: None
Stop Bits: 1
Flow Control: None

Advanced Settings

DNS Server IP: 8 . 8 . 4 . 4
Dest. Mode: Dynamic
Transfer Protocol: None
Keep Alive Time: 60 (s)
Reconnet Time: 12 (s)
Http Port: 80
UDP Group IP: 230 . 90 . 76 . 1
☐ Register Pkt: ☐ ASCII
☒ Restart for no data every 300 Sec.
☐ Enable send parameter every 5 Min.

More Advanced Settings...

Framing Rule
Max Frame Length: 1300 (Byte)
Max Interval(Smaller will better) 3 (Ms)

Get Default

Save As Default

Load Default

Modify Key

Firmware/Config

Restart Dev

Modify Setting

Cancel

More Advanced Settings

WIFI Settings

WIFI Work Mode: AP
AP/STA SSID:
Encrypt Type: No Encrypt
AP/STA Key:
AP Mode Channel: 4
DHCP Server: Enable
RJ45 WIFI Relay: Disable

The Advanced Functions Supported

☐ IO Port Control
☒ UDP Group
☒ Multi-Dest IP
☐ Proxy Function
☐ SNMP Function
☐ P2P Function

RS485 Multi-Host Support Settings

Modbus Gateway Type: Simple Modbus TCP to RTU
☐ Enable RS485 Multi-Host
Maximum wait time of RS485 query command for 0 ms(0~8191)
☐ Enable RS485 bus conflict detection
Send data only when RS485 bus is idle for 0 ms

Multi Dest-IP And Port

IP Address or DNS name	Port	Type
	0	Client Dest.
	0	Client Dest.
	0	
	0	
	0	
	0	

Function Selection

☐ Modify setting will need key
☐ Enable receive broadcast
☐ Enable P2P
☐ Send MAC when TCP establish
☐ Detect net using Ping
☐ No clear serial buffer when link
☐ Need key when TCP establish

OK

Cancel

ZLAN webpage&code download tool

Webpage directly download mode

Webpage directly in local PC:

C:\web

Special configs:

ZLMB config

MQTT config

JSON config

Reg packet

Clear all

Code file download mode

Select code file:

C:\firmware.bin

Download through the network

Device IP address or domain:

192.168.1.200

Download port (Don't modify):

1092

Download through serial port

Serial port:

Baudrate:

115200

Device modual/type:

2003

Flash size:

256

KB

DevID:

2877FE5529D4

Bind ID

Please close the opened webpage of the modual in the browser, before start download.

Download

2. Default settings for Felicity inverter / home assistant

 **WAVESHARE**
share awesome hardware

[Logout](#) [Chinese](#)

Device Information

Device Name	WSDEV0001	Firmware Version	V1.452	Device MAC	28-77-FE-55-29-D4
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Network Settings

Device IP	192.168.50.200	Device Port	502	Device Web Port	80
Work Mode	TCP Server	Subnet Mask	255.255.255.0	Gateway	192.168.50.1
Destination IP/DNS	192.168.50.200	Destination Port	502	IP mode	Static

Serial Settings

Baud Rate	2400	Databits	8	Parity	None
Stopbits	1	Flow control	None		

Advanced Settings

No-Data-Restart	Enable	No Data Restart Time	300 second	5-1270	Reconnect-time	12	1-255 second
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Multi-Host Settings

Protocol	Modbus TCP to RTU	Instruction Time out	256	32-8000ms	Enable Multi-host	Yes
RS485 Conflict Time Gap	20 5-255ms					

NOTE: 1. Multi-host is always enabled when Protocol is Modbus TCP to RTU 2. Time out is always 0 when Multi-host is disabled.
3. Time out only can be set as multply of 32.

Modify Web Login Key

New Key		Input Key Again			
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Submit

3. Establish connection Felicity/Home assistant

- A. Install HACS on HA
- B. Install Felicity via
- C. Check whether the HACS receives data
- D. If not, swap the RS485 wires in the Waveshare devices 485A and 485B ports.
- E. Use modbus poll to test connection

4. Configure Home assistant

4.1. Configuration Forecast.Solar

Install Forecast.Solar

× **Instellingen** ?

Met deze waarden kan het resultaat van Forecast.Solar worden aangepast. Raadpleeg de documentatie als een veld onduidelijk is.

API-sleutel

Declinatie (0 = Horizontaal, 90 = Verticaal)

Azimut (360 graden, 0 = Noord, 90 = Oost, 180 = Zuid, 270 = West)*

180

↕

Totaal Wattpiekvermogen van je zonnepanelen*

6120

↕

Dempingsfactor: pas de resultaten aan in de morgen

0

↕

Dempingsfactor: pas de resultaten aan in de avond

0

↕

Omvormergrootte (Watt)

10000

↕

Verzenden

4.2. Configuration Nordpool

Via HACS (**not** via normal store) install nordpool.

Nord Pool

region *
NL

Munteenheid

☐ DKK

☒ EUR

☐ NOK

☐ SEK

Selecteer valuta om prijzen in weer te geven. EUR is de basisvaluta.

☐ VAT

precision
3

low_price_cutoff
1

☐ price_in_cents

price_type

☐ MWh

☐ Wh

☒ kWh

additional_costs

Verzenden

Turn-off VAT.

In additional costs field fill in:

Supplier	Value Additional_costs field
Frank Energie	$\{\{ (0.12268 + 0.018 + (\text{current_price} * 0.21)) \mid \text{float} \}\}$
Tibber	$\{\{ (0.12268 + 0.02481 + (\text{current_price} * 0.21)) \mid \text{float} \}\}$

Double check the numbers in the contract of your energy supplier.

4.3. Finalize configuration Felecity app

× Instellingen?

register_set *
Basic (Essential sensors, fastest, low overhead) ▾

update_interval*
10 ▴ ▾

nordpool_entity

 nordpool_kwh_nl_eur_3_10_0
Hal

×

▾

nordpool_override

Selecteer een entiteit ▾

forecast_entity

 Geschatte energieproductie - vandaag
tuin ▸ Solar production forecast

×

▾

forecast_entity_tomorrow

 Geschatte energieproductie - morgen
tuin ▸ Solar production forecast

×

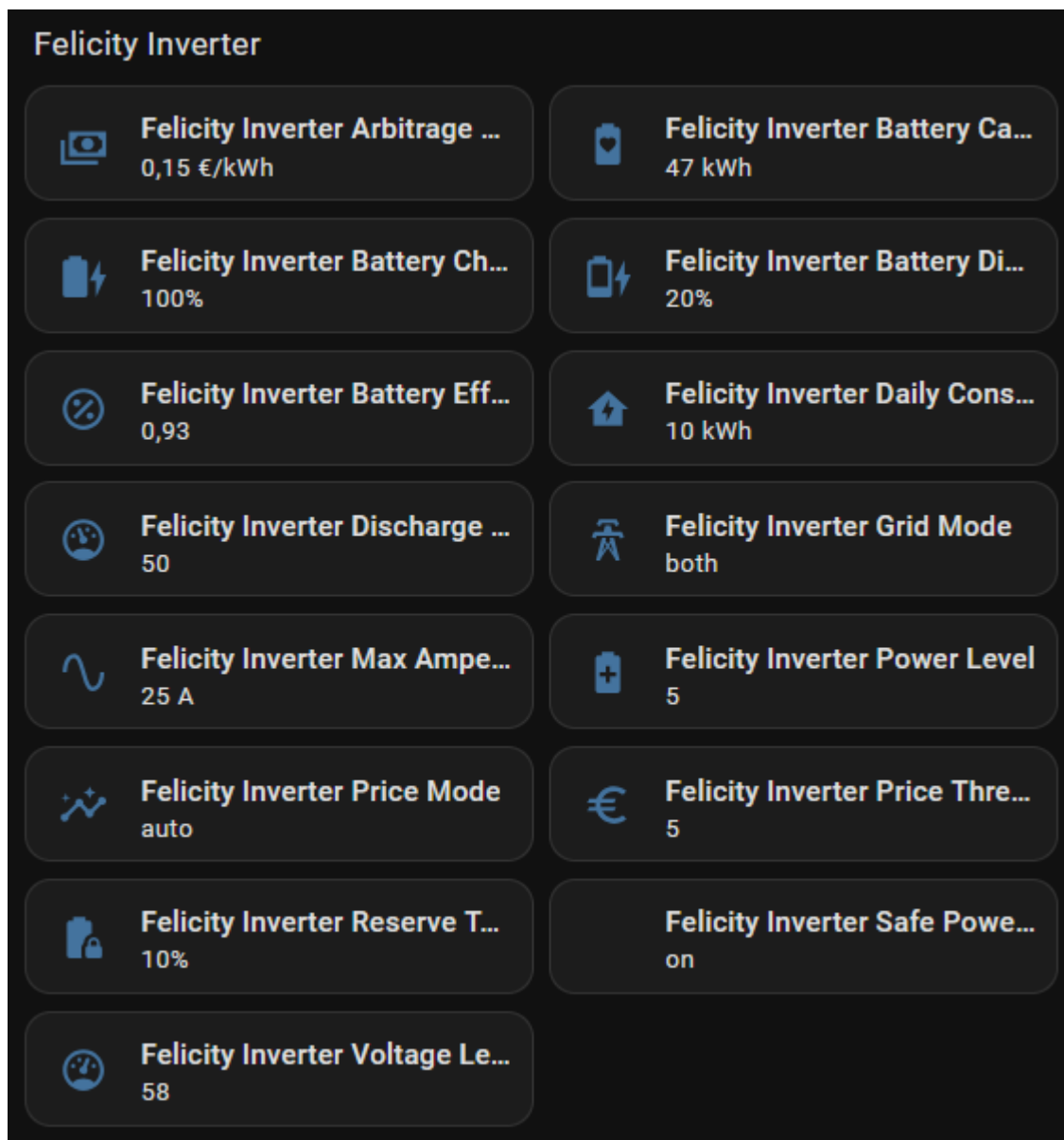
▾

consumption_override_entity

Selecteer een entiteit ▾

Verzenden

Setup done with Sebastiaan and Peter



4.4. Felicity dashboard

Add 2 cards to the dashboard. Follow instructions on: https://github.com/partach/ha_felicity

